



Topics for Today

0. Mean Filters: Recap

Spatial Filtering

I. Edge Detection

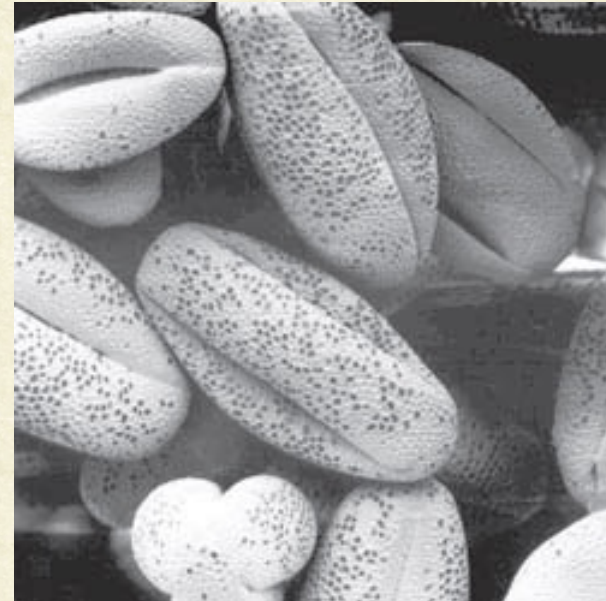
II. Image Sharpening

III. Linear Shift-Invariant Systems



CS7.404: Digital Image Processing

Monsoon 2025: **0**: Spatial Filtering: Recap
Mean Filters



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Mean / Average Filter

Note: Coefficients sum to 1

I

H

→ 3 x 3

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Weight Mask /
Kernel /
Filter

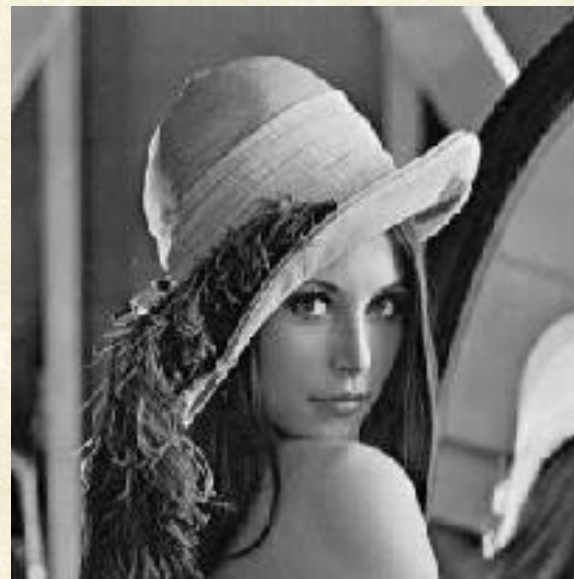
$$I'(u, v) \leftarrow \frac{1}{9} \cdot \sum_{j=-1}^1 \sum_{i=-1}^1 I(u + i, v + j)$$

$$I'(u, v) \leftarrow \sum_{j=-1}^1 \sum_{i=-1}^1 I(u + i, v + j) \cdot \boxed{H(i, j)}$$



Effect of Mask Size

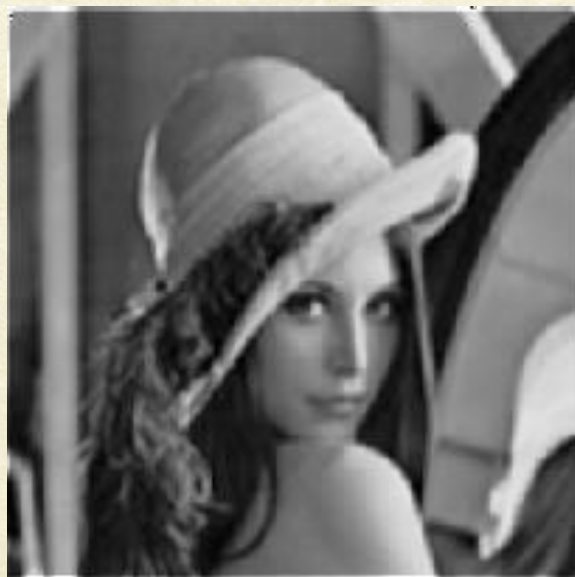
Original Image



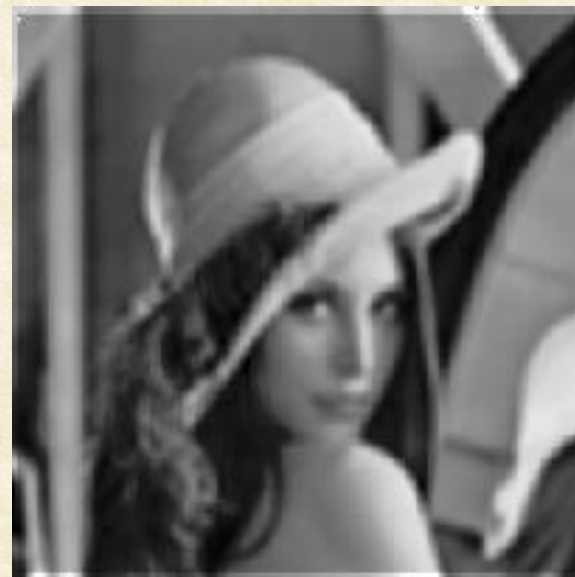
[3x3]



[5x5]



[7x7]

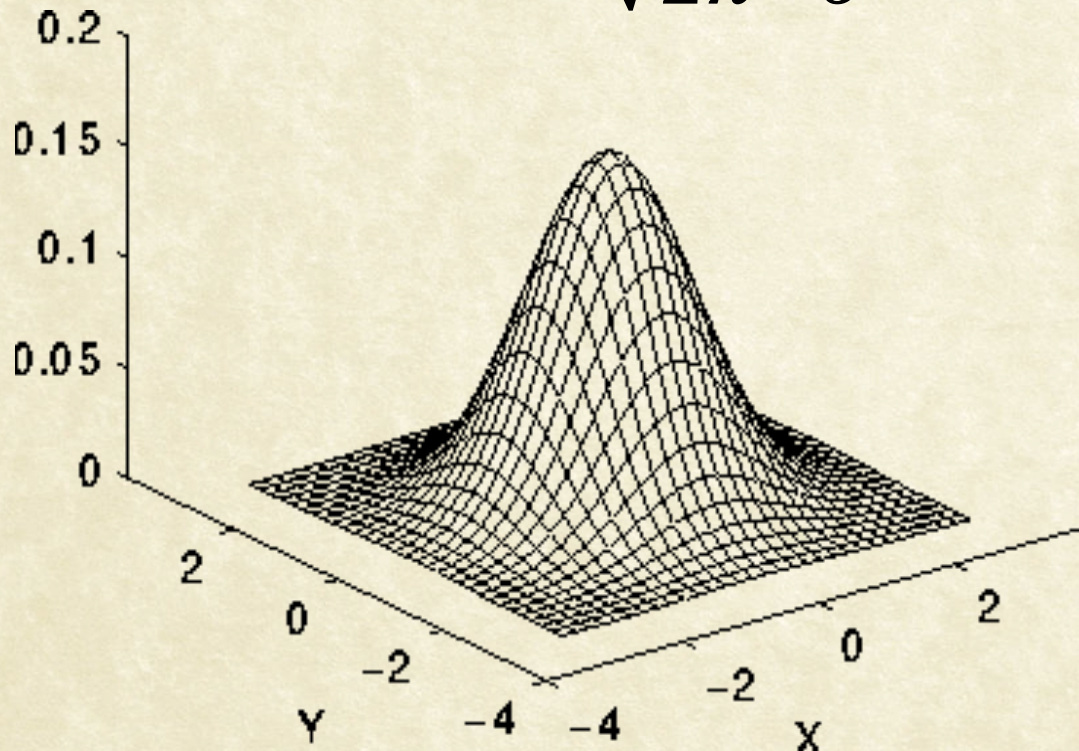




Gaussian Smoothing

- ▶ Mask weights are samples of a zero-mean 2-D Gaussian

$$G(x, y) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



$$\frac{1}{265}$$

1	4	6	4	1
4	16	26	16	4
6	26	43	26	6
4	16	26	16	4
1	4	6	4	1

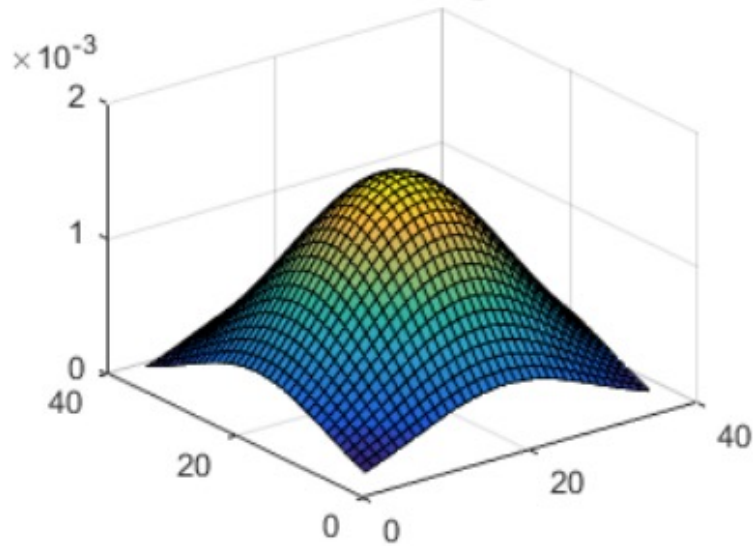
5×5 Gaussian filter, $\sigma = 1$



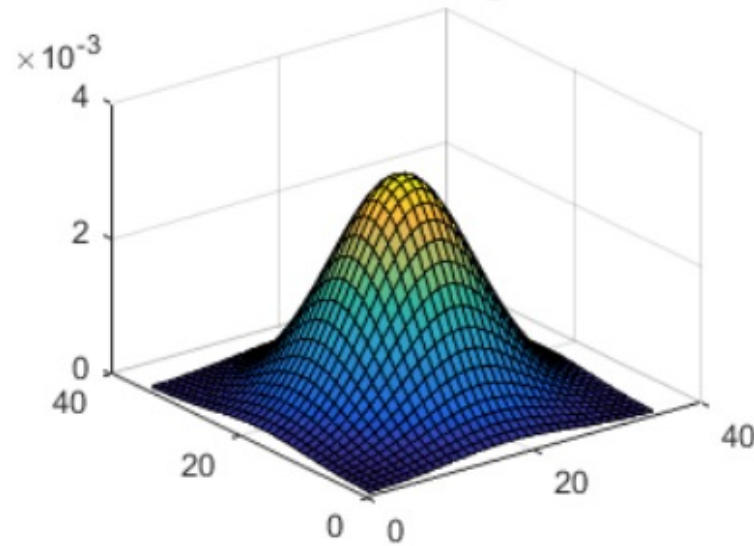
Gaussian Smoothing – Effect of sigma

$$G(x, y) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$

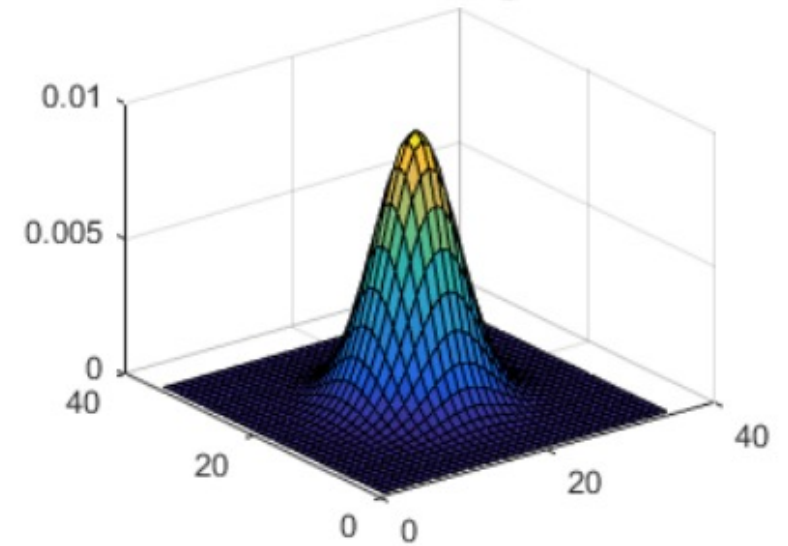
filter size = 35, sigma = 11



filter size = 35, sigma = 7



filter size = 35, sigma = 2





Gaussian Smoothing – Effect of sigma



Original Image
(Sigma 0)

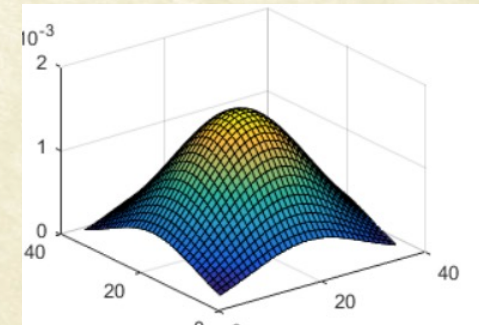
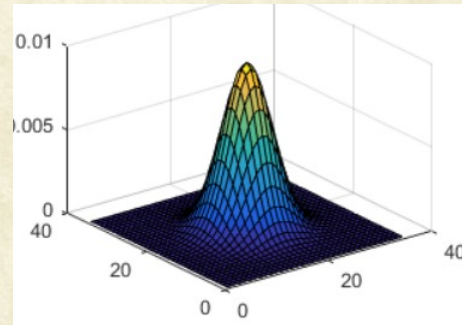


Gaussian Blur
(Sigma 0.7)



Gaussian Blur
(Sigma 2.8)

$$G(x, y) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$





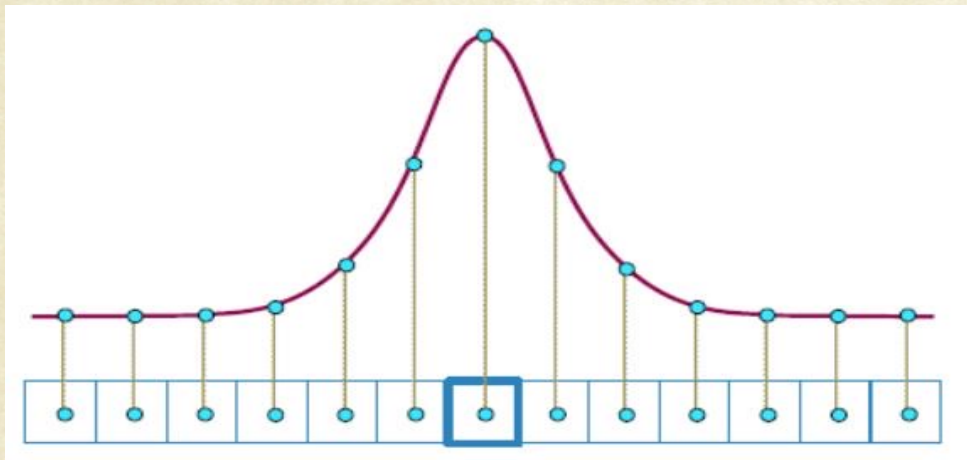
Computing Gaussian filter coefficients

Index N	Coefficients	Sum=2 ^N
0	1	1
1	1 1	2
2	1 2 1	4
3	1 3 3 1	8
4	1 4 6 4 1	16
5	1 5 10 10 5 1	32
6	1 6 15 20 15 6 1	64
7	1 7 21 35 35 21 7 1	128
8	1 8 28 56 70 56 28 8 1	256
9	1 9 36 84 126 126 84 36 9 1	512
10	1 10 45 120 210 252 210 120 45 10 1	1024

e.g., $s = 7 \times 7$

1	6	15	20	15	6	1
6	36	90	120	90	36	6
15	90	225	300	225	90	15
20	120	300	400	300	120	20
15	90	225	300	225	90	15
6	36	90	120	90	36	6
1	6	15	20	15	6	1

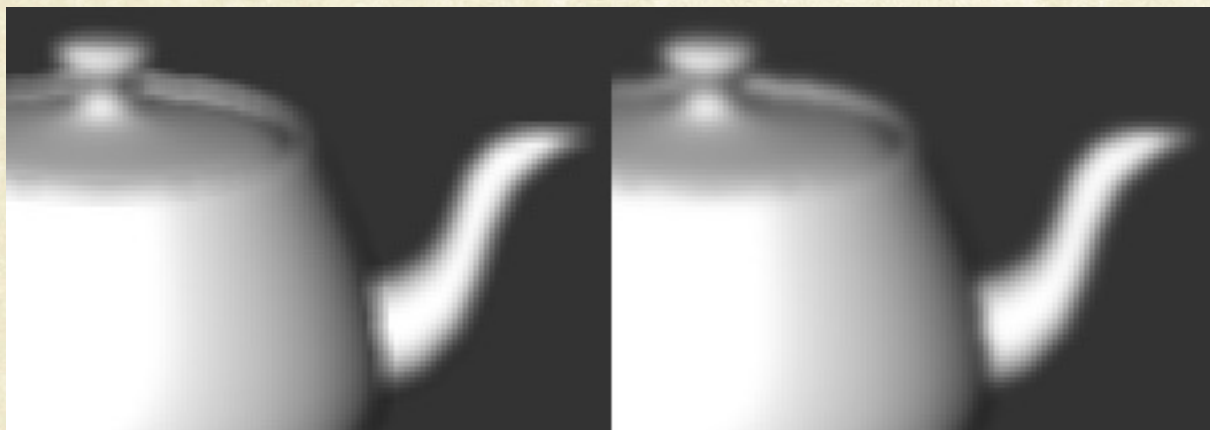
$$\frac{1}{4096} \times$$



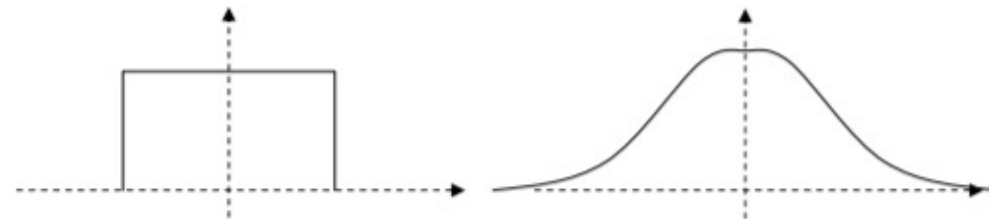
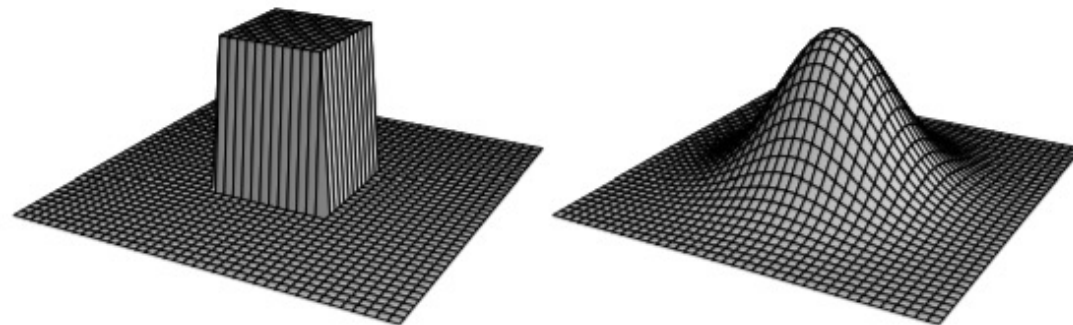
$$g_{\sigma}(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} e^{-\left(\frac{x^2}{2\sigma^2}\right)}$$



Averaging vs. Gaussian Filter



Smoother intensity transitions



0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

(a)

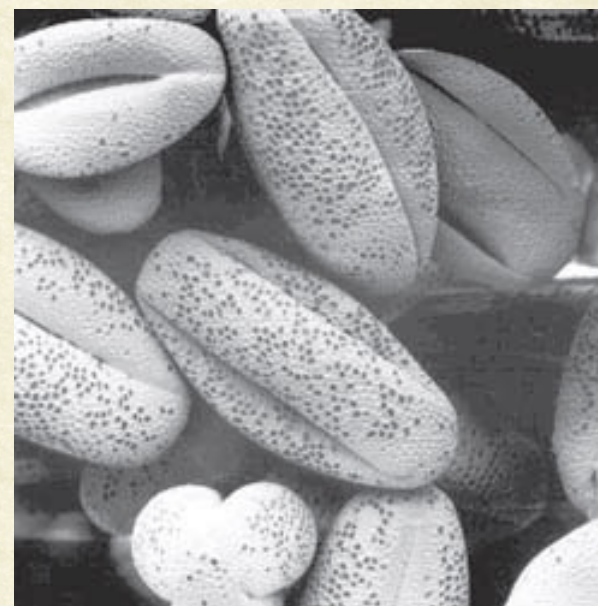
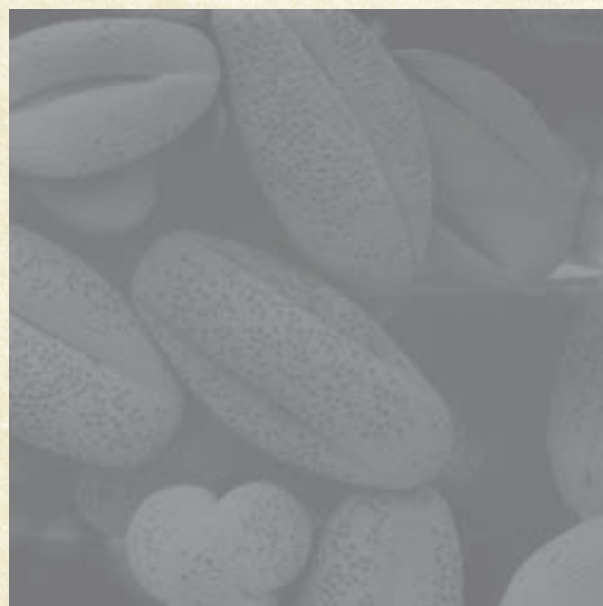
0	1	2	1	0
1	3	5	3	1
2	5	9	5	2
1	3	5	3	1
0	1	2	1	0

(b)



CS7.404: Digital Image Processing

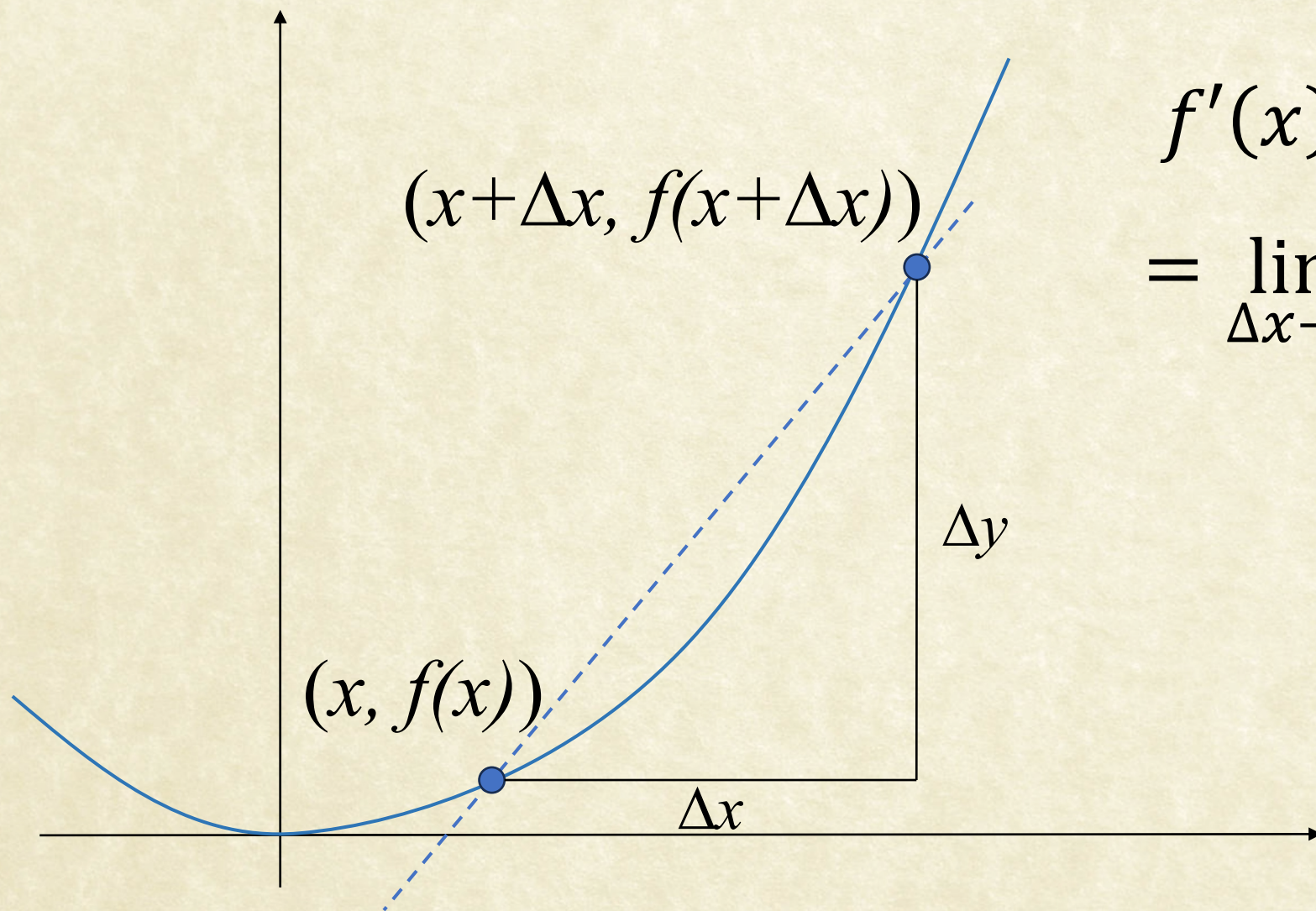
Monsoon 2025: **I**: Spatial Filtering
Gradients and Edge Detection



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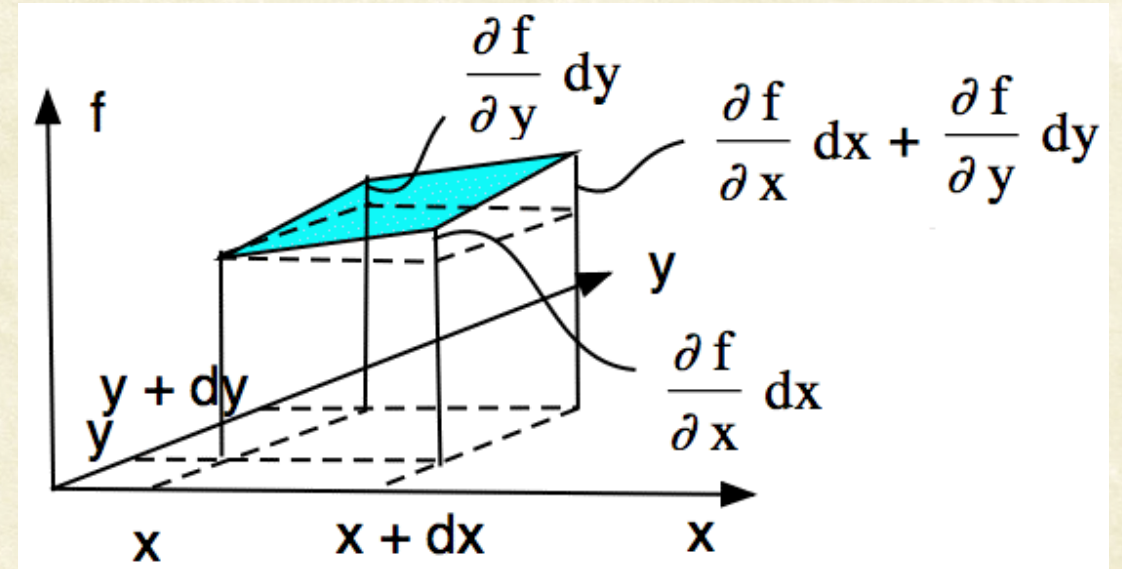
Recap: Derivatives



$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$



Recap: Derivatives



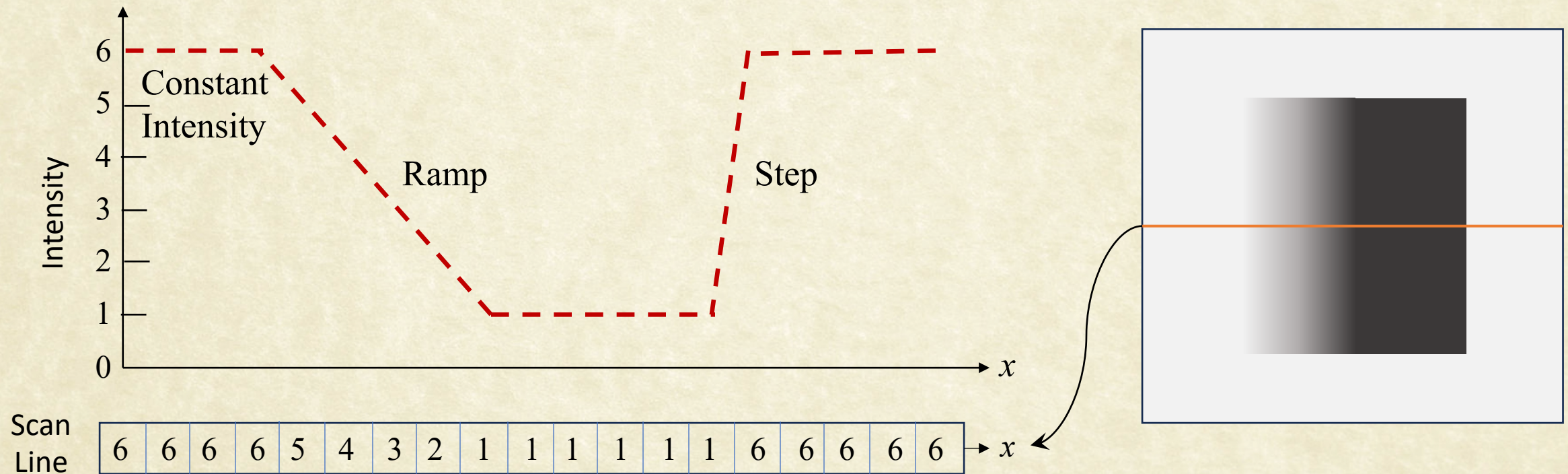
$$\frac{\partial f(x, y)}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\frac{\partial f(x, y)}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$



First Derivative: Digital Approximation

$$\frac{\partial f(x, y)}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \approx f[x + 1, y] - f[x, y]$$





Second Derivative: Digital Approximation

$$\frac{\partial f(x, y)}{\partial x} \approx f[x + 1, y] - f[x, y]$$

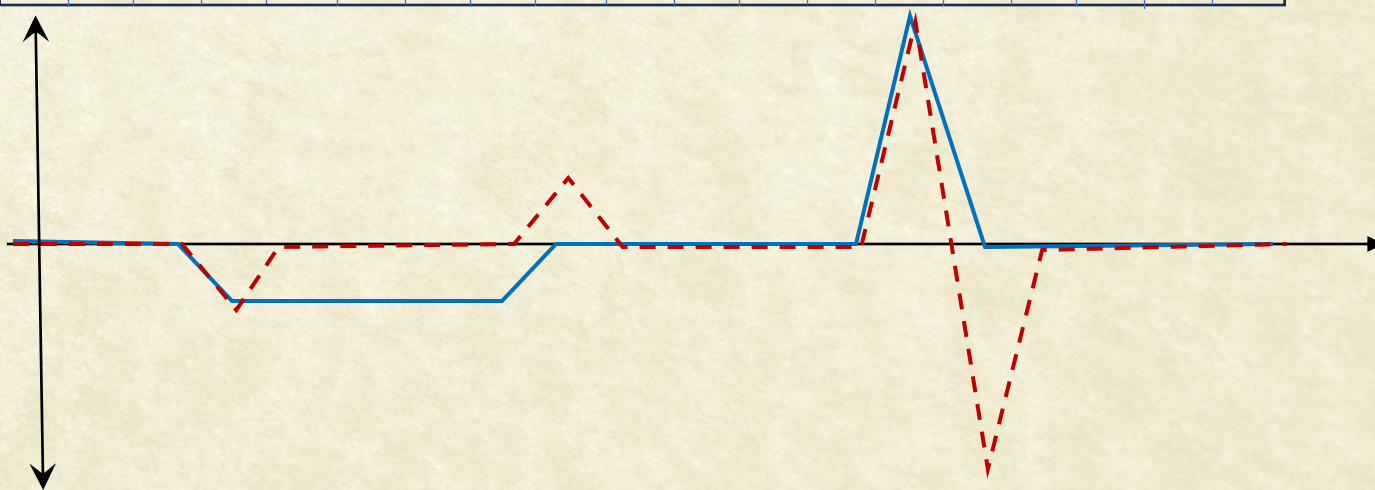
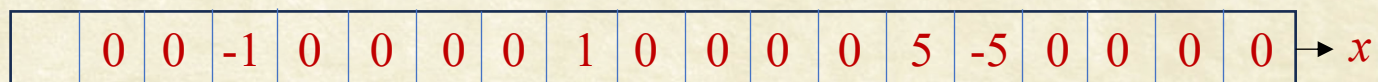
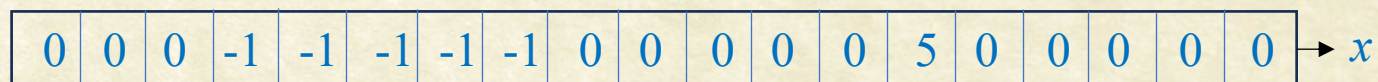
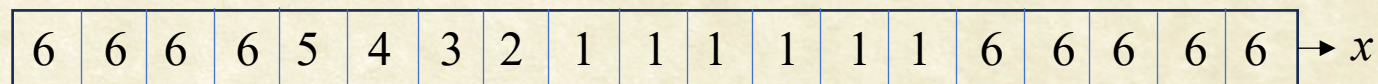
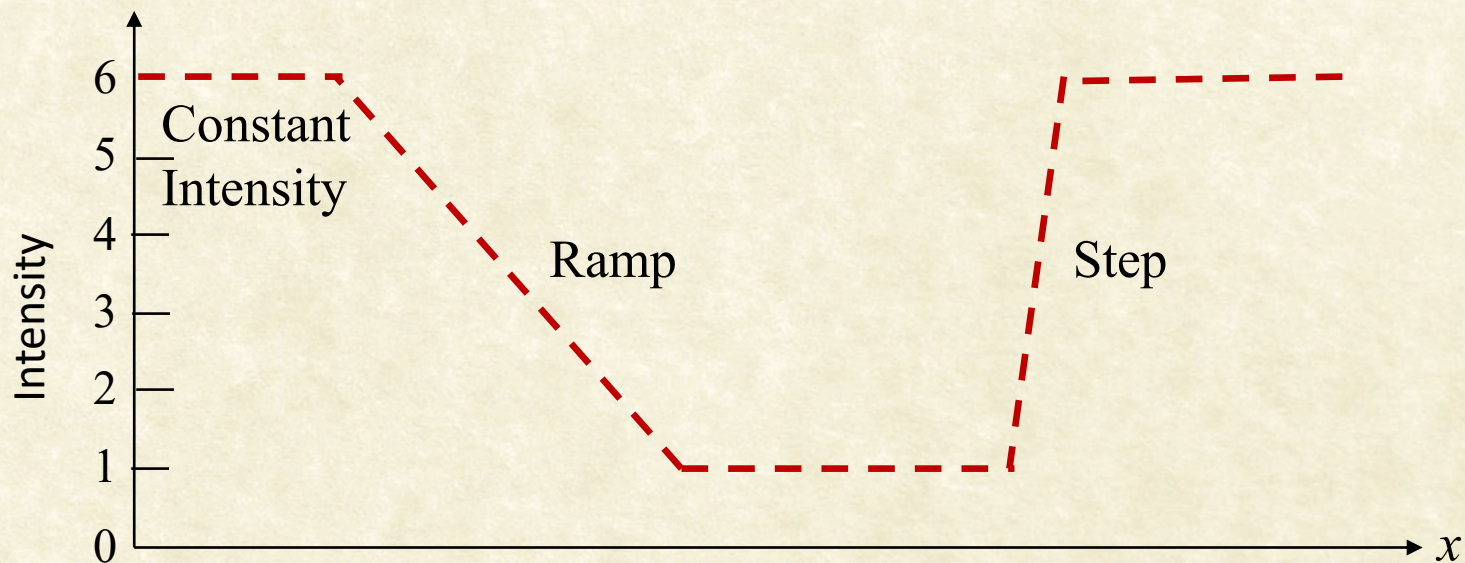
$$\frac{\partial^2 f(x, y)}{\partial x^2} \approx$$

$$(f[x + 1, y] - f[x, y]) - (f[x, y] - f[x - 1, y])$$

Scan
Line

1st derivative

2nd derivative





Alt: First Derivative as Symmetric Difference

$$\frac{\partial f(x, y)}{\partial x} \approx f[x + 1, y] - f[x, y]$$

$$\frac{\partial f(x, y)}{\partial x} \approx \frac{f[x + h, y] - f[x - h, y]}{2h}$$



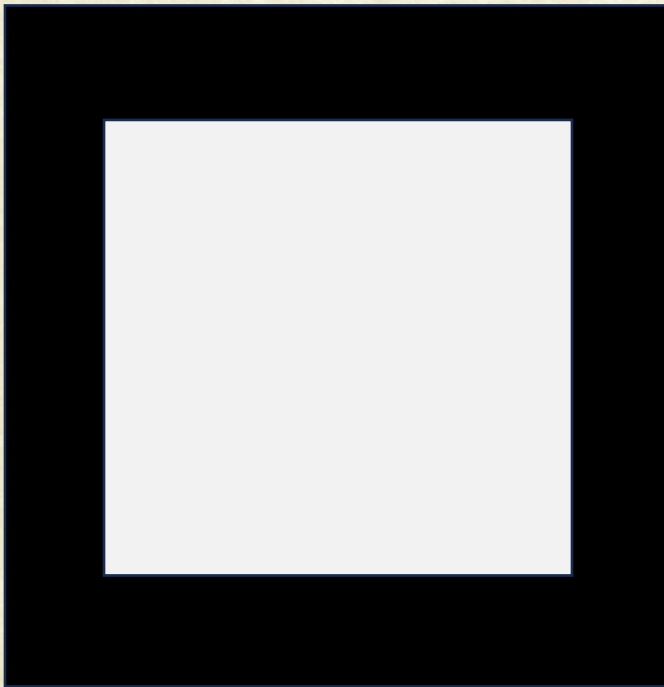
First Derivatives and Edges

$$\frac{f[x + h, y] - f[x - h, y]}{2h}$$

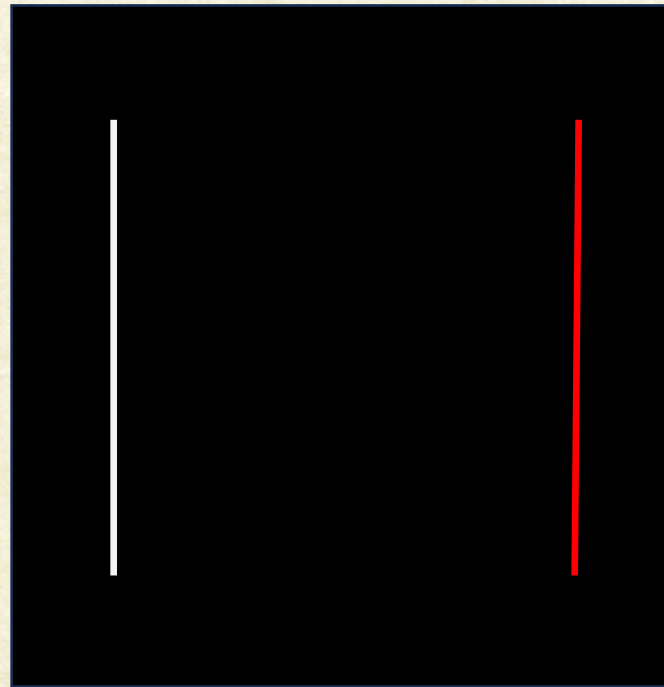
$$\rightarrow \frac{1}{2} \begin{bmatrix} -1 & 0 & 1 \end{bmatrix}$$

$$\frac{f[x, y + h] - f[x, y - h]}{2h}$$

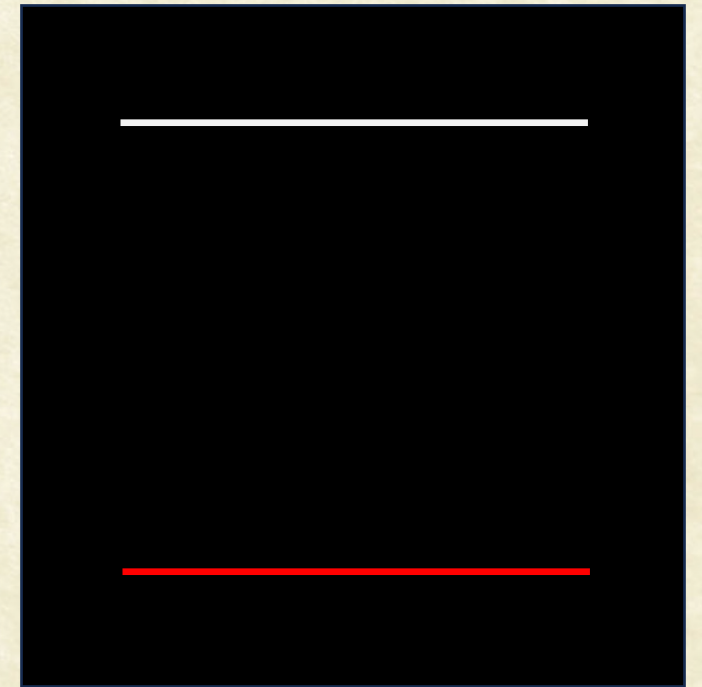
$$\rightarrow \frac{1}{2} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$



Image



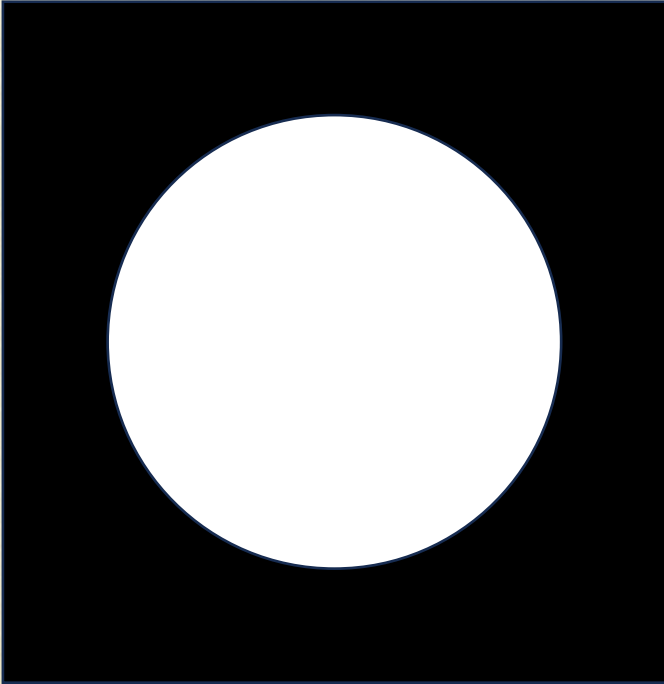
x-derivative



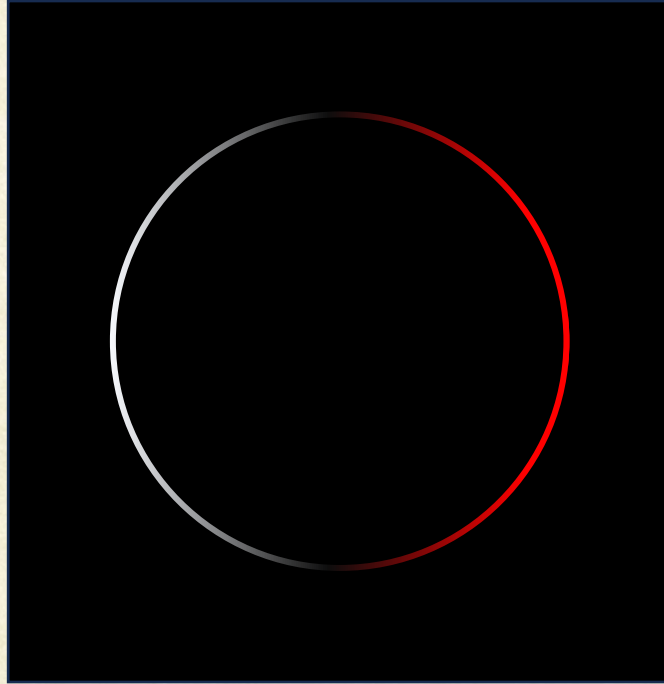
y-derivative



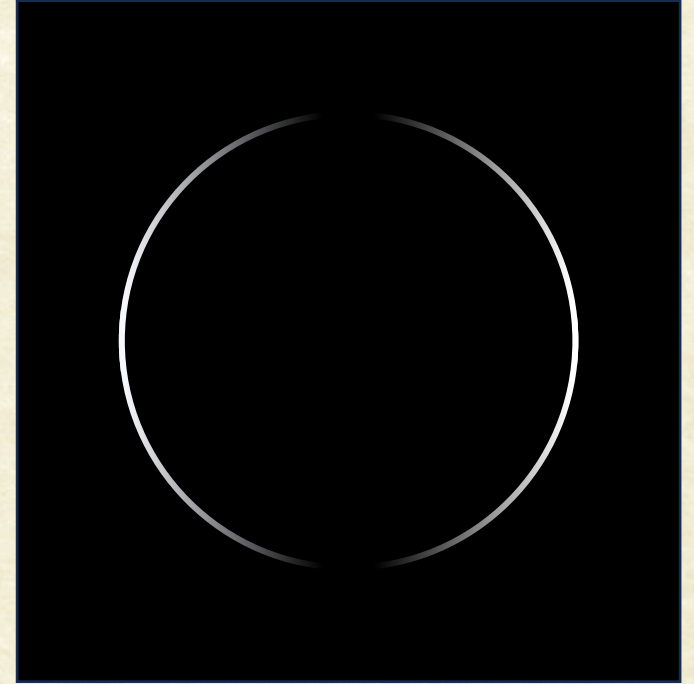
Edge 'Image'



Image



Horizontal
Derivative Image

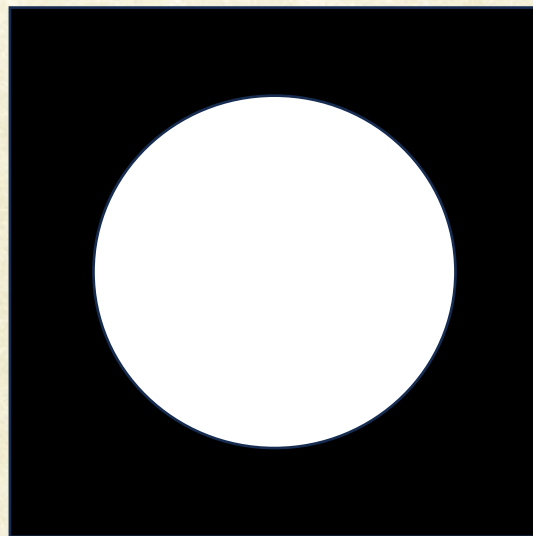


Horizontal
Edge Image



Edge 'Image'

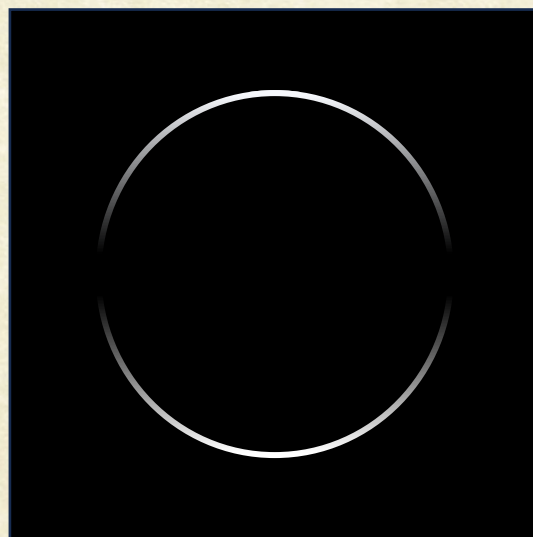
Image



Horizontal
Edge Image



Vertical
Edge Image



Edge Image

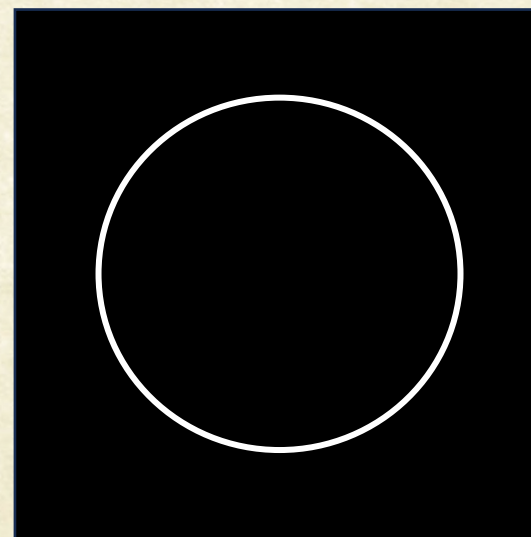


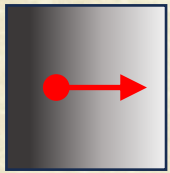


Image Gradient

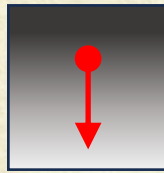
The gradient of an image:

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

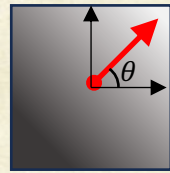
The gradient points in the direction of most rapid change (increase) in intensity.



$$\nabla f = \left[\frac{\partial f}{\partial x}, 0 \right]$$

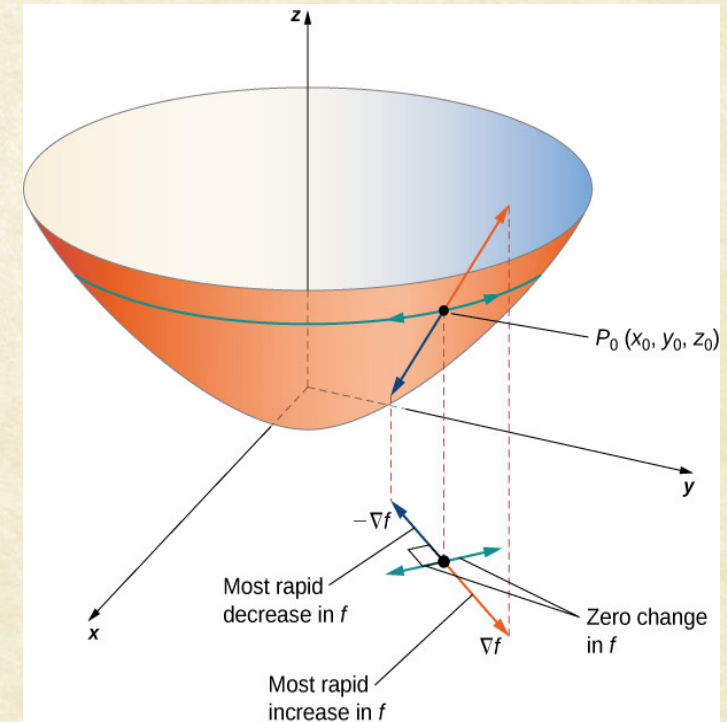
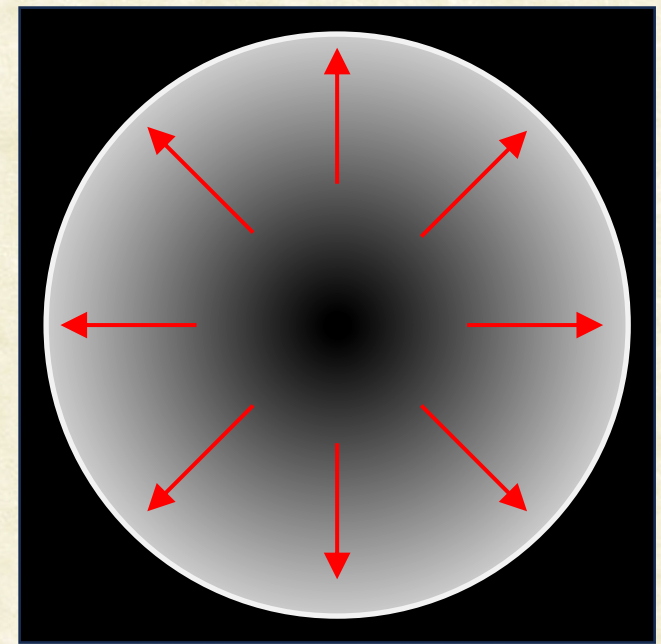


$$\nabla f = \left[0, \frac{\partial f}{\partial y} \right]$$



$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$$

$$\theta = \tan^{-1} \left(\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}} \right)$$





Prewitt Edge Filter



Dr. Judith M Prewitt

-1	0	+1
-1	0	+1
-1	0	+1

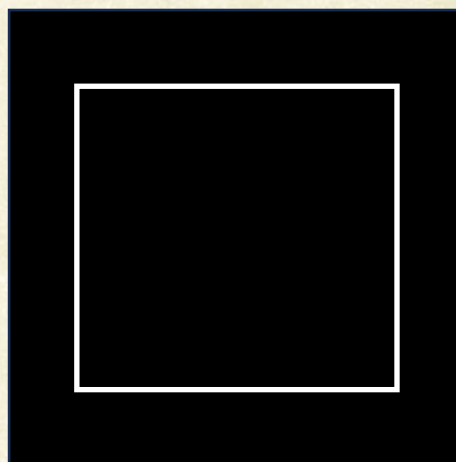
G_x

+1	+1	+1
0	0	0
-1	-1	-1

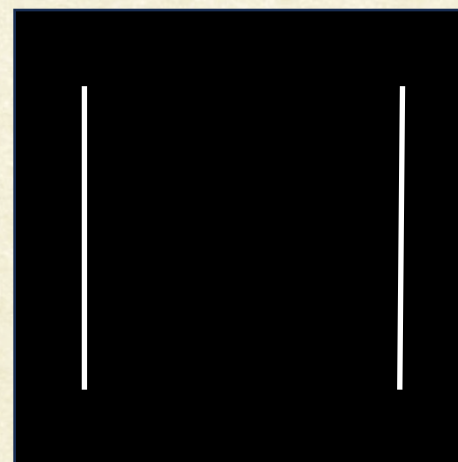
G_y



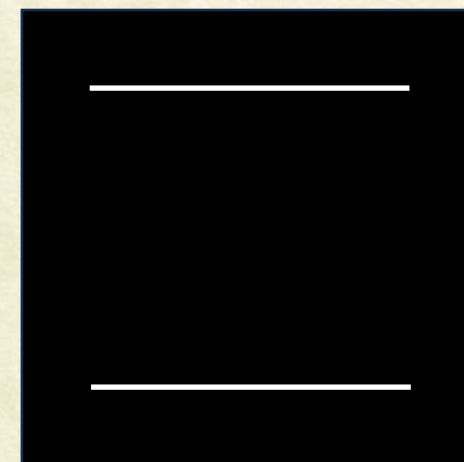
Edge is Perpendicular to Gradient



Image



Gradient in x



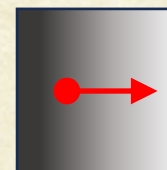
Gradient in y

-1	0	+1
-1	0	+1
-1	0	+1

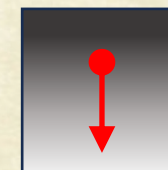
G_x

+1	+1	+1
0	0	0
-1	-1	-1

G_y



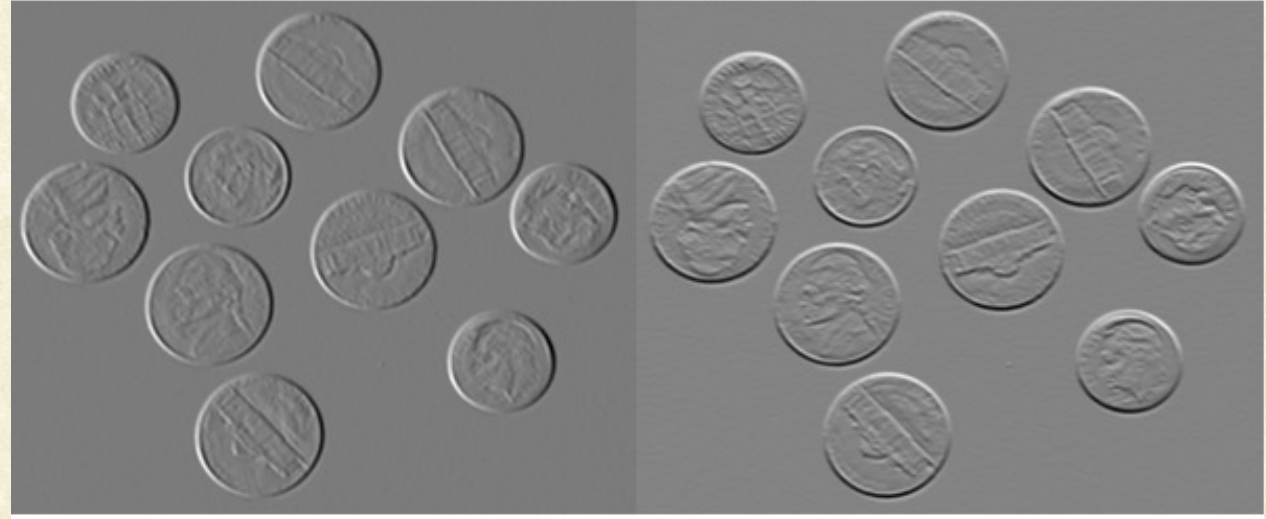
$$\nabla f = \left[\frac{\partial f}{\partial x}, 0 \right]$$



$$\nabla f = \left[0, \frac{\partial f}{\partial y} \right]$$



Edge is Perpendicular to Gradient

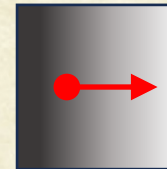


-1	0	+1
-1	0	+1
-1	0	+1

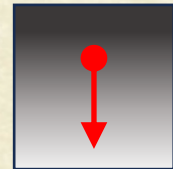
G_x

+1	+1	+1
0	0	0
-1	-1	-1

G_y



$$\nabla f = \left[\frac{\partial f}{\partial x}, 0 \right]$$

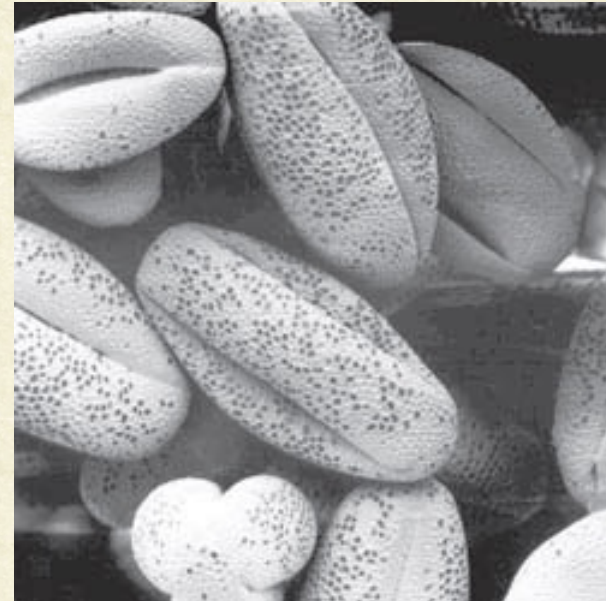


$$\nabla f = \left[0, \frac{\partial f}{\partial y} \right]$$



CS7.404: Digital Image Processing

Monsoon 2025: II: Spatial Filtering
Image Sharpening



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2-D Laplacian Filter

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

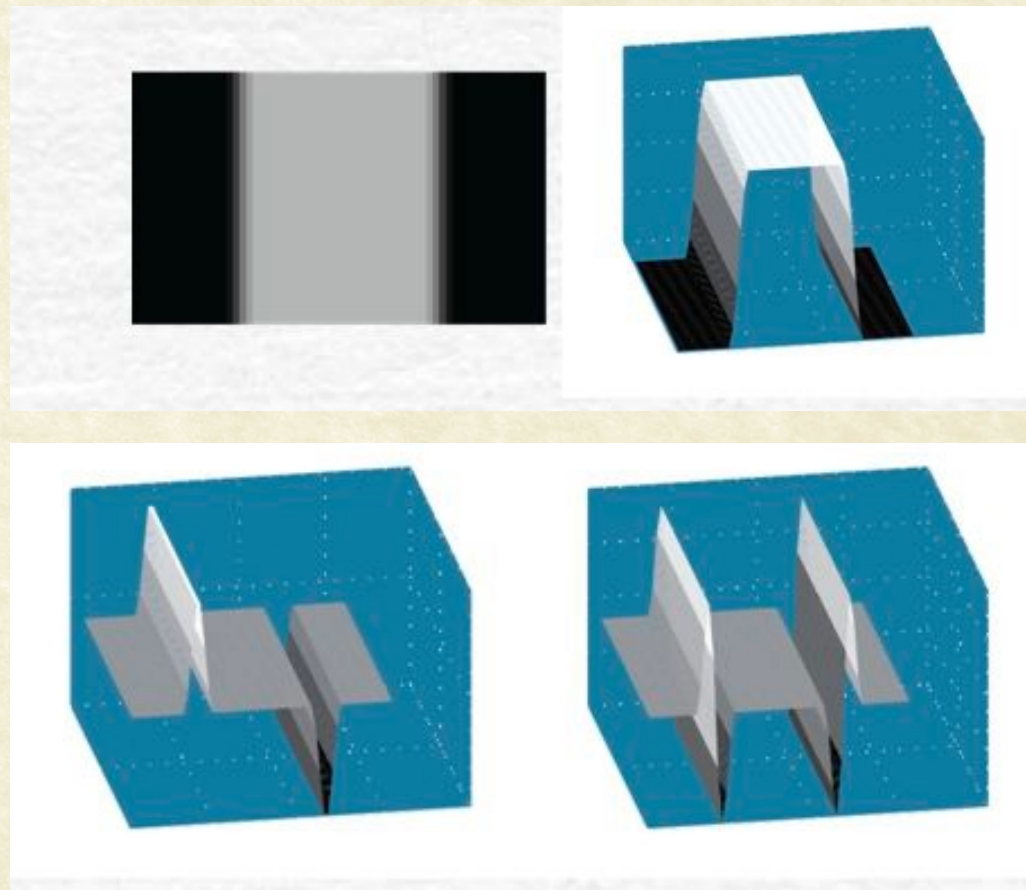
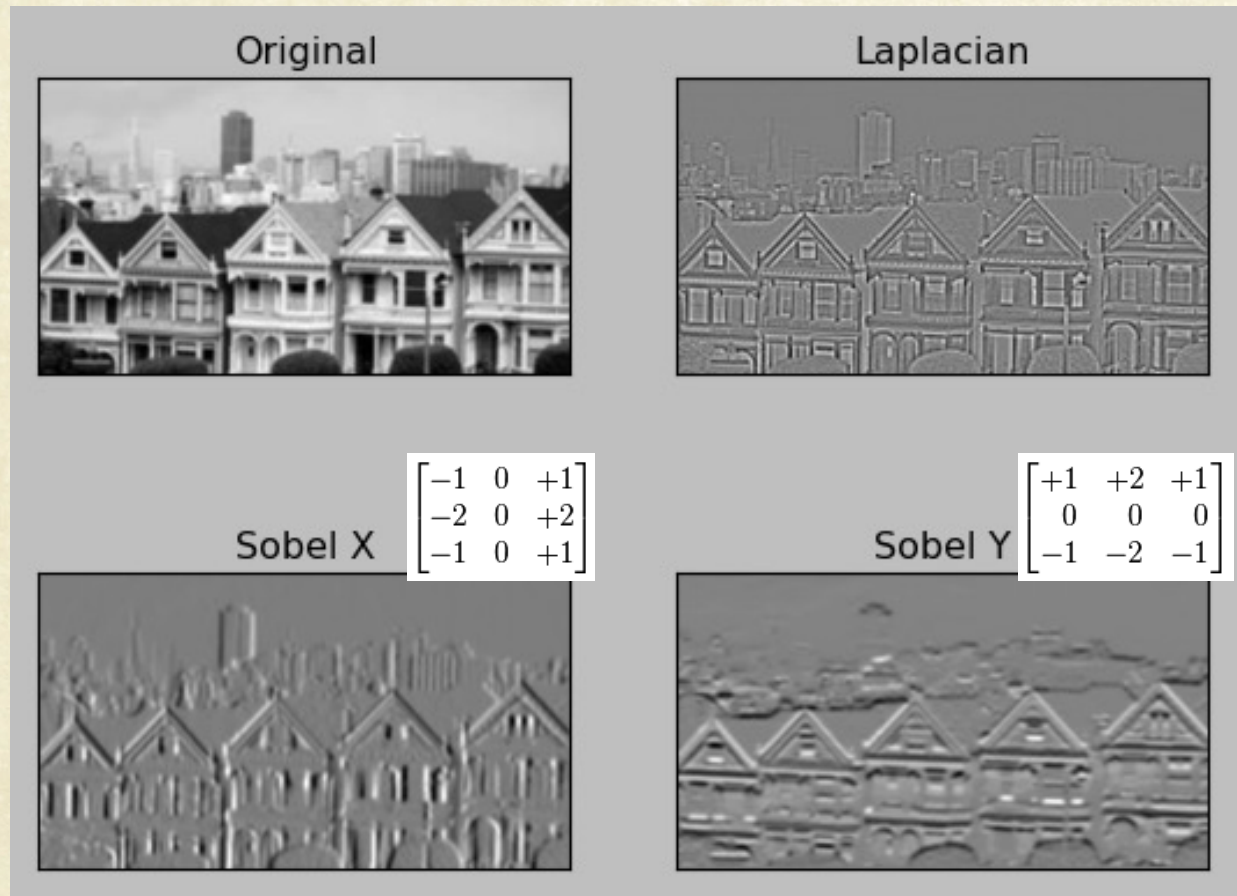
0	1	0
1	-4	1
0	1	0



Edge Masks: Sobel, Laplacian

0	1	0
1	-4	1
0	1	0

Note:
Coefficients
sum to 0



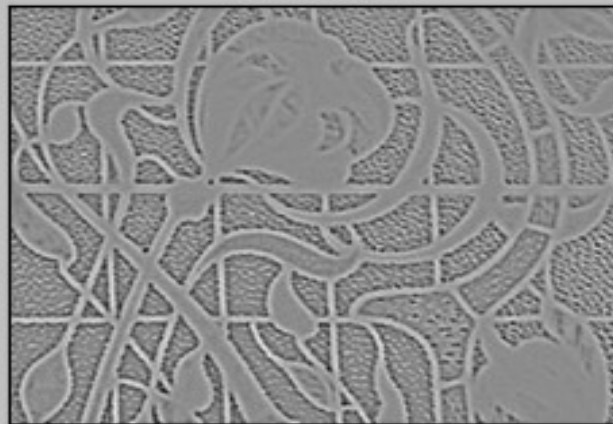


Edge Masks: Sobel, Laplacian

Original



Laplacian



Sobel X



Sobel Y



$$\begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}$$

$$\begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

0	1	0
1	-4	1
0	1	0

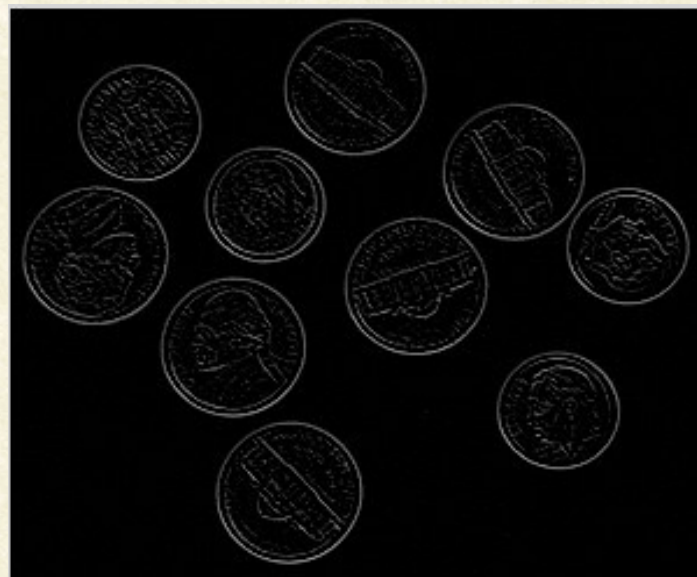


Image Sharpening

$$I(u, v)$$



$$\nabla^2 I(u, v)$$

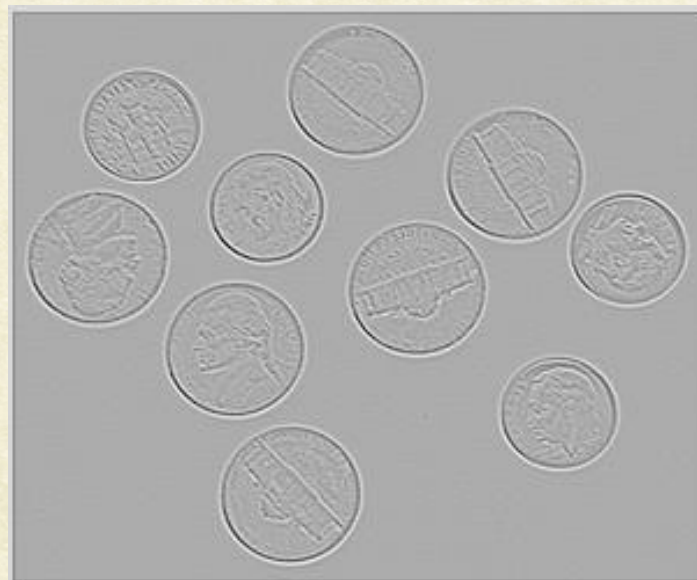


$$I'(u, v) = I(u, v) + \nabla^2 I(u, v)$$



$$\nabla^2 I(u, v) + 128$$

(For visualization)



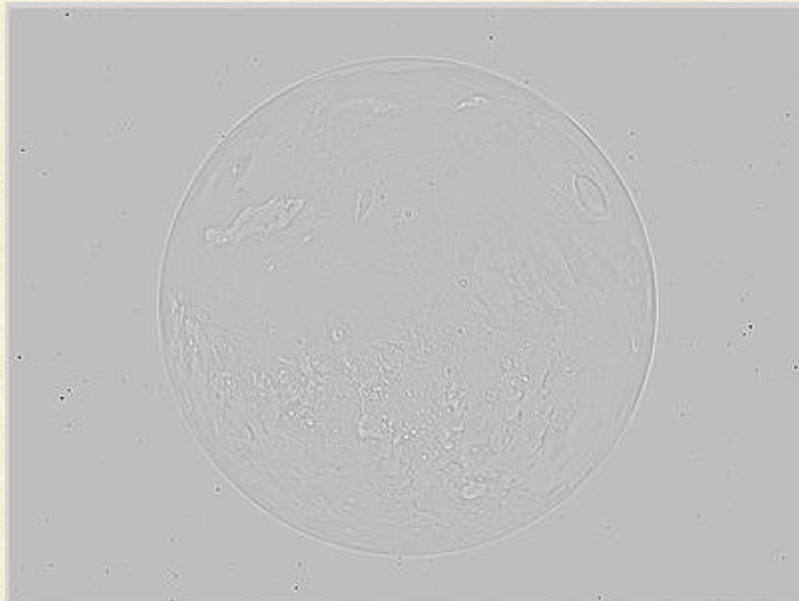


Sharpening (Unsharp Masking)

$$I(u, v)$$



$$\nabla^2 I(u, v)$$



$$I'(u, v) = I(u, v) + \nabla^2 I(u, v)$$

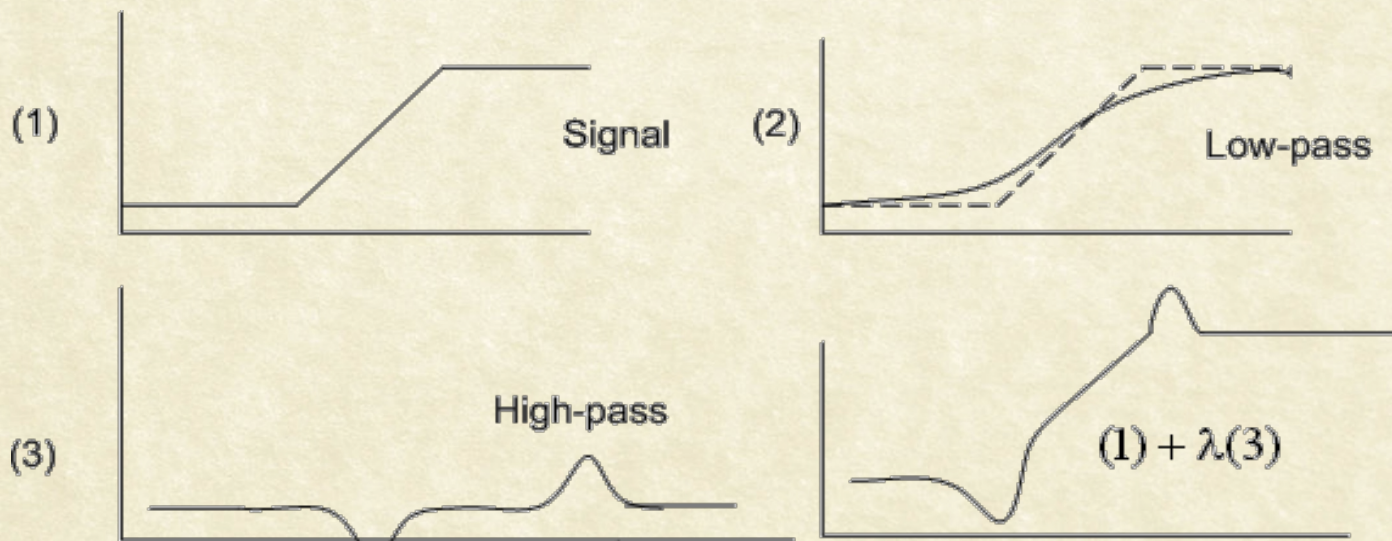


Image Courtesy: NASA



Image Sharpening: Example



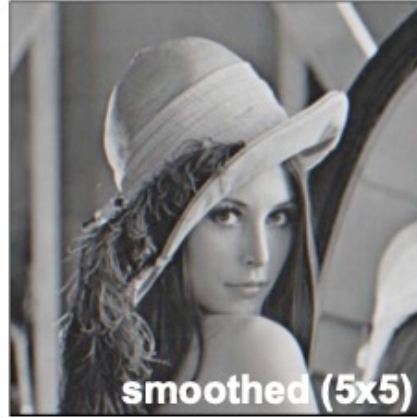


Highboost Filtering

- What does blurring take away?



—



=



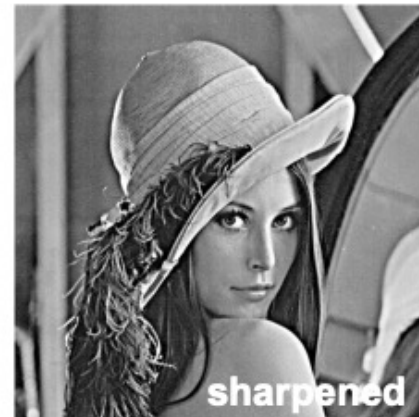
- Let's add it back:



+ a



=





Unsharp Masking vs Highboost Filtering





USM vs. HBF as Spatial Filters

A=1			A=2		
W = 9A-1			W = 17		
-1	-1	-1	-1	-1	-1
-1	W	-1	-1	17	-1
-1	-1	-1	-1	-1	-1

- ▶ If $A=1$, we get unsharp masking. $I'(u, v) = I(u, v) + \nabla^2 I(u, v)$
- ▶ If $A>1$, original image is added back to detail image (HBF).



Corner Cases: Padding

$M = 3$

For each valid location $[x,y]$ in S

$a \leftarrow$ Average of intensities in a $M \times M$ neighborhood centered on $[x,y]$

$D[x,y] = \text{round}(a)$

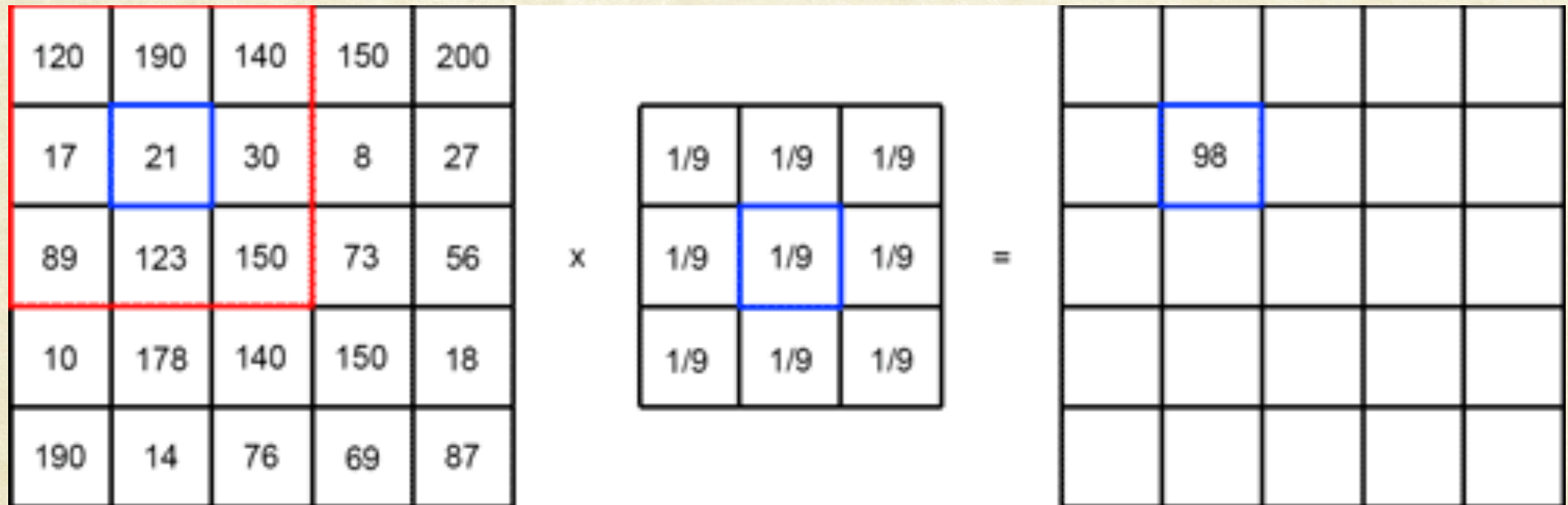
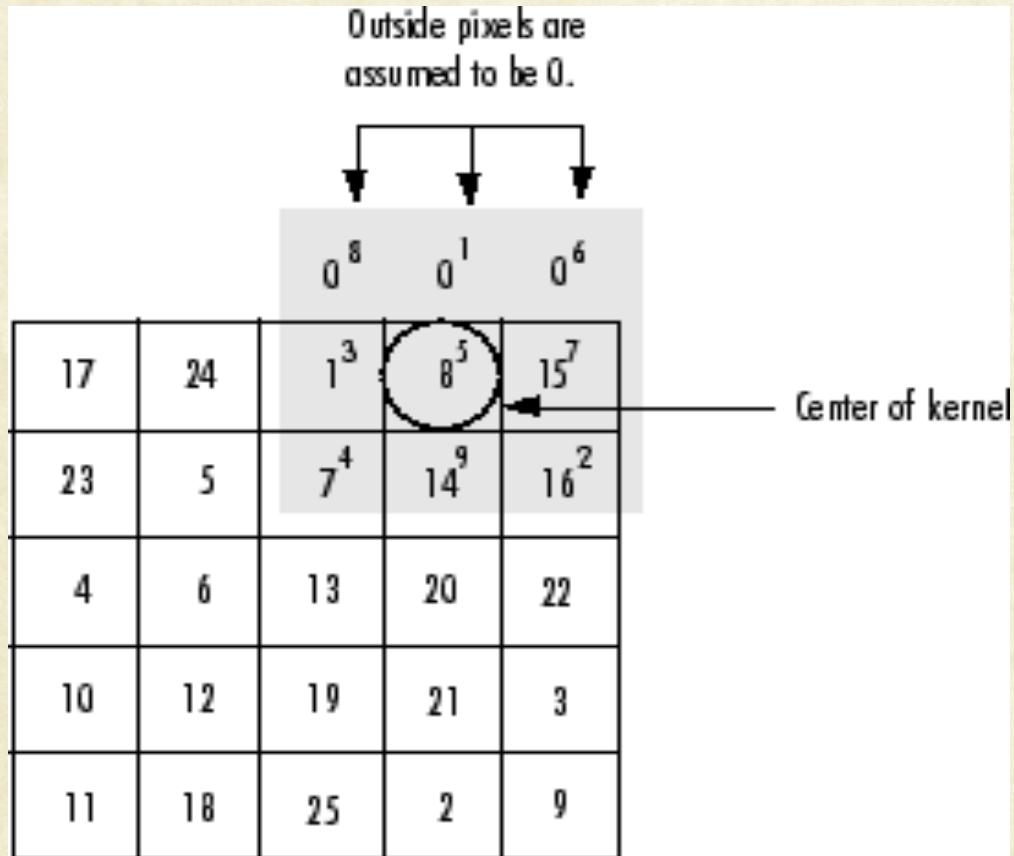
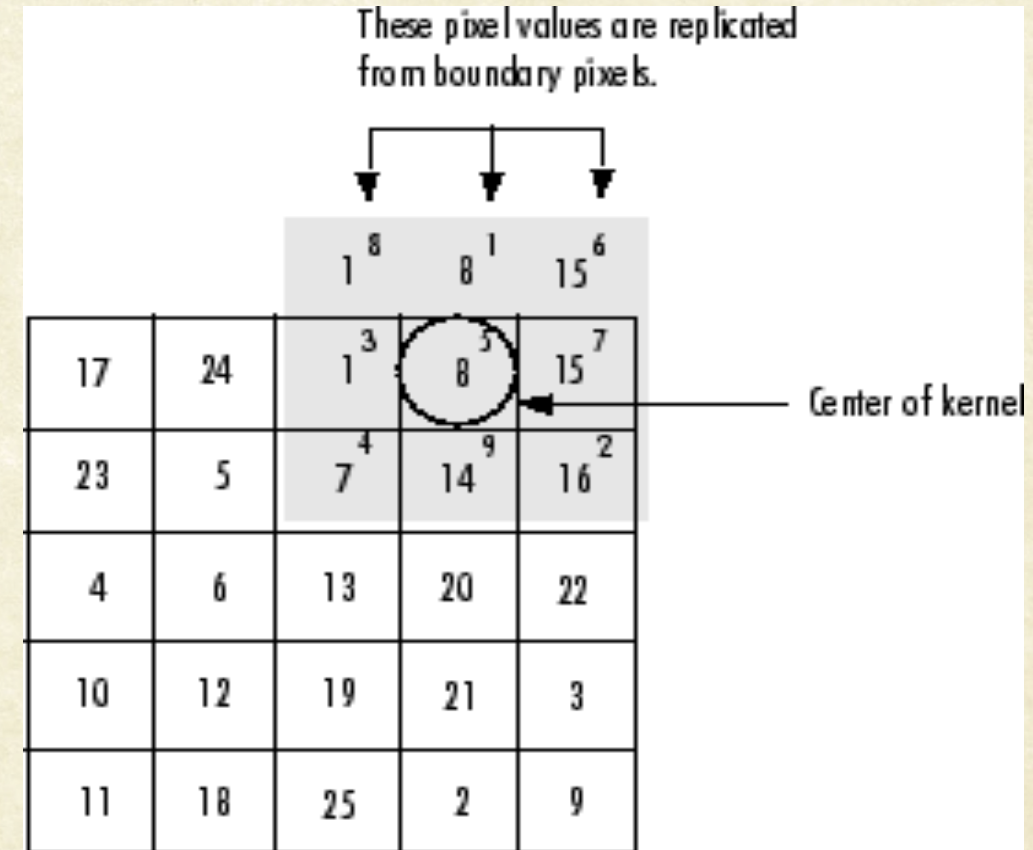




Image Padding



Zero/Constant



Replicate/Reflect

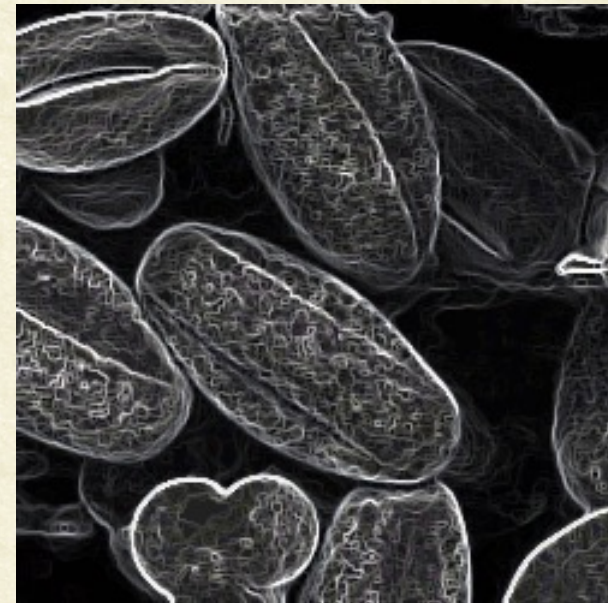
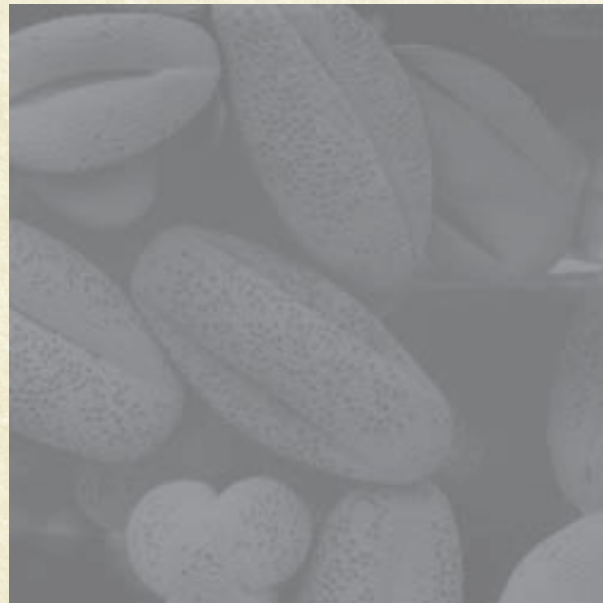


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: **III**: Linear Shift Invariant Systems

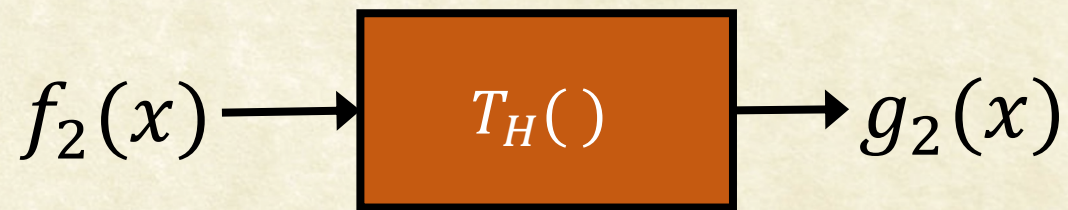
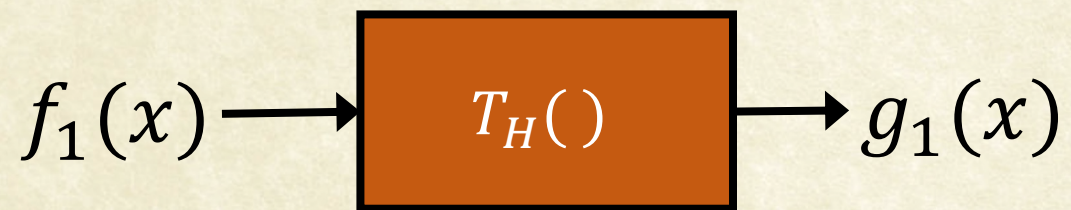


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Linear Systems

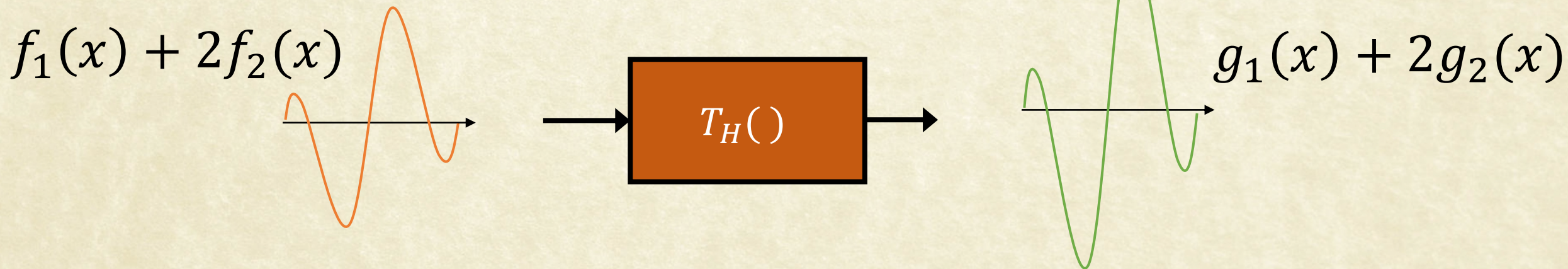
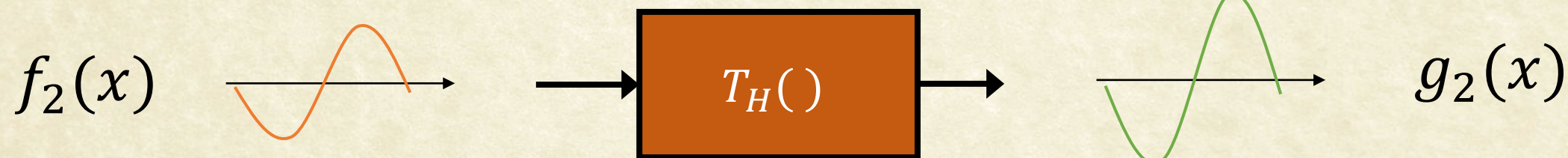
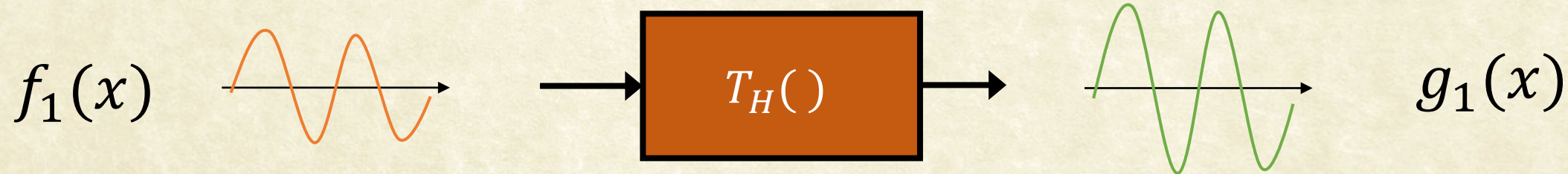
$$g(x) = T_H(f(x))$$





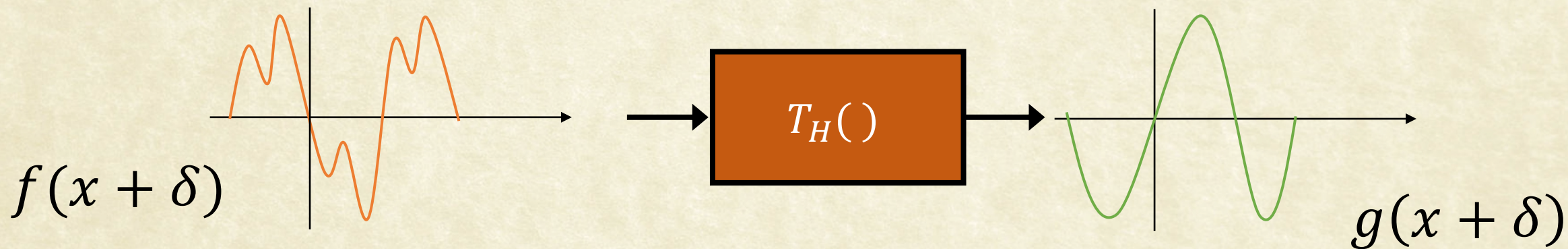
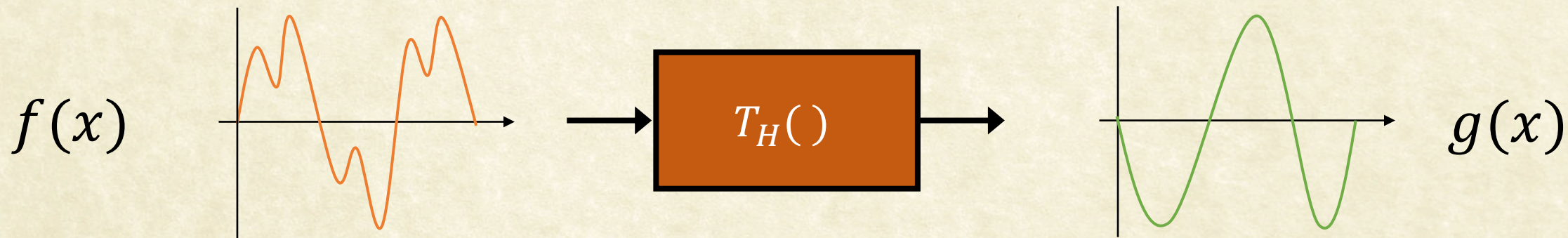
Linear Systems

$$g(x) = T_H(f(x))$$





Shift Invariant Systems





Questions?